NOVA SCHOOL OF SCIENCE & TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE

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Master in Computer Science and Engineering

COST AND OPERATIONS OPTIMIZATION IN MULTI-CLOUD ENVIRONMENTS OVER GENAI FRAMEWORKS

MASTER IN COMPUTER SCIENCE AND ENGINEERING

SPECIALIZATION IN SPECIALIZATION NAME

NOVA University Lisbon

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Abstract

Keywords: Multi-Cloud · Generative AI · Cost Optimisation · Cloud Operations · GenAI Frameworks

Resumo

Palavras-chave: Multi-Cloud · Inteligência Artificial Generativa · Otimização de Custos · Operações em Nuvem · Frameworks GenAI

Disclosure of Delegation to Generative Artificial Intelligence

Duarte José Lourenço Belo declares[1] the use of Generative AI in the research work and writing process of this document. According to the GAIDeT taxonomy [6], the following tasks were delegated to Generative AI tools under full human supervision:

Conceptualization

- [x] Idea generation
- [x] Defining the research objective
- [x] Formulating research questions and hypotheses
- [ ] Feasibility assessment and risk evaluation
- [ ] Preliminary hypothesis testing

Literature Review

- [x] Literature search and systematization
- [x] Writing the literature review
- [ ] Analysis of market trends and/or patent environment
- [ ] Evaluation of novelty and identification of gaps

Methodology

- [x] Research design
- [x] Development of experimental or research protocols
- [ ] Selection of research methods

Software Development and Automation

- [x] Code generation
- [x] Code optimization
- [ ] Process automation
- [ ] Creation of algorithms for data analysis

Data Management

- [x] Data collection
- [x] Validation
- [x] Data cleaning
- [ ] Data curation and organization
- [ ] Data analysis
- [ ] Visualization
- [ ] Reproducibility testing

[1] This declaration was generated using LuaLaTeX [7] and the LaTeX package `aidisclose` [4].

Writing and Editing

- [x] Text generation
- [x] Proofreading and editing
- [x] Summarizing text
- [ ] Formulation of conclusions
- [ ] Adapting and adjusting emotional tone
- [ ] Translation
- [ ] Reformatting
- [ ] Press releases and outreach materials

Ethics Review

- [x] Bias analysis and discrimination assessment
- [x] Ethical risk analysis
- [x] Monitoring compliance with ethical standards
- [ ] Data confidentiality monitoring

Supervision

- [ ] Quality assessment
- [ ] Trend identification
- [ ] Identification of limitations
- [ ] Recommendations
- [ ] Publication support

Generative AI tools used: ChatGPT-5.2, Gemini 3 Pro, Claude 4.5 Opus, and Perplexity 12.12.25.

Additional comment #1: The AI was used primarily for refining the code in Section 3, but the algorithm logic was derived manually.

Additional comment #2: This is a second additional comment!

Additional comment: This is a third additional comment! It is starred and, thus, does not get a number. Also, this comment is longer and, hopefully, will span to the next line!

Additional comment #3: This is the third non-starred additional comment!

Contents

Abstract i

Resumo ii

Disclosure of Delegation to Generative Artificial Intelligence iii

Contents v

List of Figures vii

List of Tables viii

1 Introduction 1

1.1 Context and Motivation 1

1.2 Problem Statement 1

1.3 Research Objectives 1

1.4 Contributions 1

1.5 Document Structure 1

2 Background 2

2.1 Foundations of Neural Language Representation 2

2.1.1 Distributed Word Representations 2

2.1.2 The Transformer Architecture 3

2.1.3 Multi-Cloud Environments 4

2.1.4 Cost Optimisation in Cloud Computing 4

2.2 Related Work 4

2.3 Summary 4

3 Proposed Approach 5

3.1 Overview 5

3.2 Architecture 5

3.3 Components and Design Decisions 5

3.4 Cost and Operations Model 5

3.5 Summary 5

4 Implementation **6**
4.1 Overview 6
4.2 Technology Stack 6
4.3 Framework Implementation 6
4.3.1 Component A 6
4.3.2 Component B 6
4.4 Deployment and Configuration 6
4.5 Summary 6

5 Evaluation **7**
5.1 Evaluation Methodology 7
5.2 Experimental Setup 7
5.3 Results 7
5.4 Discussion 7
5.5 Summary 7

6 Conclusion **8**
6.1 Summary of Contributions 8
6.2 Limitations 8
6.3 Future Work 8
6.4 Final Remarks 8

Bibliography **9**

Appendices

A Appendix Title **10**

B Appendix Title **11**

List of Figures

List of Tables

Listings

1

Introduction

1.1 Context and Motivation

1.2 Problem Statement

1.3 Research Objectives

1.4 Contributions

1.5 Document Structure

2

Background

2.1 Foundations of Neural Language Representation

The ability to represent language in a form that machines can process and reason over is a prerequisite for any modern Generative AI system. Two landmark contributions established the theoretical and practical foundations upon which contemporary large language models (LLMs) and retrieval-augmented architectures are built: the introduction of dense word representations via neural networks, and the development of the Transformer architecture based solely on attention mechanisms.

2.1.1 Distributed Word Representations

Prior to 2013, natural language processing relied predominantly on sparse, high-dimensional representations such as one-hot encodings, which offered no intrinsic notion of semantic similarity between words. Mikolov et al. [5] addressed this limitation by proposing Word2Vec, a computationally efficient framework for learning continuous vector representations of words from large unlabelled corpora. The model is trained through a self-supervised prediction task using one of two architectures: Continuous Bag-of-Words (CBOW), which predicts a target word from its surrounding context, or Skip-gram, which performs the inverse prediction. The resulting word embeddings encode both syntactic and semantic structure in a dense vector space, enabling linear algebraic operations over meaning — most notably the well-known relationship $\vec{king} - \vec{man} + \vec{woman} \approx \vec{queen}$.

The significance of Word2Vec extends well beyond the paper itself. By demonstrating that semantic similarity can be captured as geometric proximity in a continuous vector space, it established the distributional hypothesis as an actionable engineering principle. This insight directly underlies the design of modern embedding models and vector databases, which are central components in Retrieval-Augmented Generation (RAG) pipelines [3]. In the context of this work, the quality and cost of the embedding stage in the Maestro framework's RAG pipeline is a direct consequence of choices made in this representational paradigm.

2.1.2 The Transformer Architecture

Despite the advances in distributed representations, sequence modelling remained constrained by recurrent architectures such as LSTMs and GRUs, which process tokens sequentially and suffer from vanishing gradients over long contexts. Vaswani et al. [8] proposed a radical departure: the Transformer, an architecture that dispenses with recurrence entirely and relies solely on attention mechanisms to model dependencies between all positions in a sequence simultaneously. The core operation, scaled dot-product attention, computes output representations as a weighted sum of value vectors, where weights are derived from the compatibility between query and key projections:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

By running multiple such attention operations in parallel — Multi-Head Attention — the model learns to simultaneously attend to different types of relationships across the input, such as syntactic dependencies and semantic co-references. Positional encodings are injected additively into the input embeddings to compensate for the architecture's inherent permutation invariance. The resulting model achieved 28.4 BLEU on the WMT 2014 English-to-German translation benchmark, surpassing all prior approaches while requiring a fraction of the training compute [8].

The Transformer's parallelism over sequence positions unlocked a new scaling regime for neural language models. Architectures derived from it — most notably encoder-only models such as BERT [2] for representation learning, and decoder-only models such as the GPT series [1] for text generation — became the standard building blocks of enterprise GenAI systems. Critically, the attention mechanism scales quadratically with sequence length, $O(n^2)$, meaning that context length is a primary cost driver in token-priced LLM APIs. This architectural property is foundational to the FinOps strategies explored in this dissertation, namely prompt compression, semantic caching, and dynamic model selection, as each seeks to reduce the effective n processed by the model or the frequency with which inference is invoked at all [5, 8].

Together, these two works define the representational substrate of modern GenAI: Word2Vec established that meaning can be encoded as position in a learned vector space, and the Transformer established that context-sensitive reasoning over such representations can be performed efficiently and at scale. The remainder of this chapter examines the architectural patterns, operational frameworks, and cost optimisation techniques that build upon these foundations in production-grade enterprise systems.

2.1.3 Multi-Cloud Environments

2.1.4 Cost Optimisation in Cloud Computing

2.2 Related Work

2.3 Summary

3

Proposed Approach

3.1 Overview

3.2 Architecture

3.3 Components and Design Decisions

3.4 Cost and Operations Model

3.5 Summary

4

Implementation

4.1 Overview

4.2 Technology Stack

4.3 Framework Implementation

4.3.1 Component A

4.3.2 Component B

4.4 Deployment and Configuration

4.5 Summary

5

Evaluation

5.1 Evaluation Methodology

5.2 Experimental Setup

5.3 Results

5.4 Discussion

5.5 Summary

6

Conclusion

6.1 Summary of Contributions

6.2 Limitations

6.3 Future Work

6.4 Final Remarks

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A

Appendix Title

B

Appendix Title